

Devin J. Williams

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Research Interests

My research uses large observational datasets to investigate the physical processes driving galaxy stellar mass growth and morphological evolution over cosmic time. In particular, I focus on how merger-driven growth varies among galaxy populations, and how different large-scale cosmic environments shape galaxy assembly over time. I study the evolution of galaxy stellar haloes, whose extended structure preserve signatures of past assembly, to probe the contribution from different types of mergers and environmental processes.

Education

- PhD in Astronomy, Saint Mary's University, Halifax, Canada** 09/2023 – present
- Supervisors: Dr. Ivana Damjanov, Dr. Marcin Sawicki
 - Thesis: The Role of Cosmic Environment in Galaxy Stellar Halo Evolution (exp. 08/2027)
- MSc in Astronomy, Saint Mary's University, Halifax, Canada** 09/2021 – 08/2023
- Supervisors: Dr. Ivana Damjanov, Dr. Marcin Sawicki
 - Thesis: Investigating Galaxy Size and Mass Growth Through Stellar Halo Assembly Since $z \sim 1.1$
- BSc in Astrophysics, Saint Mary's University, Halifax, Canada** 05/2019 – 05/2021
- Institutional Honors: summa cum laude
- BBA in Economics, University of New Brunswick, Fredericton, Canada** 09/2011 – 05/2015
- Minor: International Business

Research Experience

- Graduate Research Assistant (PhD), Saint Mary's University** 09/2023 – present
- Designed extensive simulation-based tests to validate galaxy image correction and photometric fitting procedures
 - Gained expertise working with telescope pipelines (hscPipe), and accessing and calibrating raw data
 - Developed Python-based code to identify quiescent cluster galaxies in photometric datasets
 - Created CLAUDS+HSC-SSP quiescent cluster galaxy sample, used in multiple publications
 - Investigated the role of cluster environments in stellar halo assembly through light and color profile analysis
 - Contributed to projects with international collaborators on galaxy size evolution and properties of BCGs
- Graduate Research Assistant (MSc), Saint Mary's University** 09/2021 – 08/2023
- Helped test & develop a parallelized Python suite for automated galaxy image analysis – GalPrime
 - Developed a forward-modelling procedure for removing PSF-related smearing effects from galaxy images
 - Extracted light profiles of $\sim 330,000$ galaxies from multi-band photometric images using HPC systems
 - Developed methodology to study stellar halo assembly via evolution in galaxy light profiles
 - Contributed to the DEUS observation proposal for the Canada-France-Hawaii Telescope (CFHT)
- Undergraduate Student Researcher (NSERC USRA), Saint Mary's University** 05/2021 – 08/2021
- Developed coding skills for planned MSc project through hands-on work and ACENET courses
 - Gained experience acquiring and analyzing large multi-band imaging datasets from CLAUDS & HSC-SSP
- Undergraduate Research Assistant (BSc), Saint Mary's University** 09/2020 – 01/2021
- Queried Gaia Satellite databases and analyzed stellar proper motion data using Python
 - Studied globular cluster dynamics and searched for high velocity stars using Gaia data

Teaching & Relevant Work Experience

- Telescope Operator & Observatory Tour Guide, Burke-Gaffney Observatory, Halifax** 09/2021 – Present
- Help undergraduate astronomy students with course-related observational projects
 - Lead observatory tours and guided observation sessions open to the public
 - Communicate astronomy concepts to a wide range of audiences through classroom-based presentations
- Graduate Teaching Assistant (MSc + PhD), Saint Mary's University** 09/2021 – present
- Grade course material for undergraduate physics and astronomy courses
 - Provide detailed feedback on students' work and assist students during labs and office hours
 - Designed & presented mini lectures on astronomy topics to 1st year physics students
- Undergraduate Teaching Assistant (BSc), Saint Mary's University** 09/2020 – 05/2021
- Graded coursework for 1st year physics courses, providing detailed feedback on students' work
 - Assisted with in-class lab experiments, guiding students through key concepts
- Operations Assistant, SunSolar Energy Technologies, QC, Canada** 08/2015 – 09/2018
- Handled inventory database management for thousands of commercial products
 - Managed business accounts across several e-commerce platforms
 - Generated product performance reports and assessed new opportunities via market analytics

Publications

- Williams, D. J. et al., *"Tracing Quiescent Galaxy Assembly Across Environments Through Stellar Halo Color Evolution"*, submitted to *ApJ*.
- A. George, I. Damjanov, M. Sawicki, S. Arnouts, L. Chen, G. Desprez, S. Gwyn, T. Moutard, M. Pizzardo, A. Sajina, D. J. Williams, et al., *"Bulge+Disk Morphology in Rest-frame UV and Optical: Size–Mass Relations Reveal Distinct Growth Paths for Star-forming and Quiescent Galaxies"*, submitted to *ApJ*
- Souchereau, H., Damjanov, I., Sawicki, M., Williams, D. J., et al. 2026, *"GalPRIME: Automated. Quality-Assessed 1-D Light Profile Extraction for Large Galaxy Surveys"*, submitted to *ApJ*
- Williams, D. J. et al. 2026, *"The Role of Cluster Environments in Quiescent Galaxy Stellar Halo Assembly"*, accepted at *ApJ*, Aug 2027, DOI: 10.3847/1538-4357/ae95fc
- Williams, D. J. et al. 2025, *"The Growth of Galaxy Stellar Haloes Over $0.2 \leq z \leq 1.1$ "*, *ApJ*, 989, 107, doi: 10.3847/1538-4357/ade9a8.
- George, A., Damjanov, I., Sawicki, M., Williams, D. J., et al. 2025, *"Effects of Environment on the Size Evolution of Quiescent Galaxies: Comparing Galaxies in Clusters and in the Field at Two Rest-frame Wavelengths"*, *ApJ*, 987, 45, doi: 10.3847/1538-4357/addc6b.

Conferences & Presentations

- "The Role of Cluster Environments in Quiescent Galaxy Stellar Halo Assembly"*, presentation, GalFirenze: The Drivers of Galaxy Evolution Through Cosmic Time, Florence, Italy, 06/2026
- "The Role of Cluster Environments in QG Stellar Halo Assembly"*, poster & presentation, BeyondHubble: Revisiting the Hubble Sequence Across Cosmic Time, Kavli Institute for Cosmology, Cambridge, U.K., 04/2026
- "Building Galaxy Outskirts: The Role of Galaxy Mergers and Cosmic Environment"*, keynote speaker, Royal Astronomical Society of Canada Event, Halifax, Canada, 02/2026
- "The Growth of Galaxy Stellar Haloes Over $0.2 \leq z \leq 1.1$ "*, poster & presentation, Canadian Astronomical Society Annual General Meeting (CASCA 2025), Halifax, Canada, 06/2025
- "The Growth of Galaxy Stellar Haloes Over $0.2 \leq z \leq 1.1$ "*, presentation, Institute for Computational Astrophysics Research Symposium, Halifax, Canada, 01/2025
- "Investigating Galaxy Assembly Through the Buildup of Stellar Haloes Since $z = 1.1$ "*, presentation, Institute for Computational Astrophysics Research Symposium, Halifax, Canada, 02/2024

Technical Skills & Certifications

Programming & Data Analysis Skills

- Python, R, Fortran for scientific computing and statistical analysis
- Bash for Unix shell scripting; HTML5/CSS for website development
- Parallel computing and large-scale analysis on HPC systems (OpenMP, Slurm)
- FITS, Jupyter Notebooks, Conda, SQL, ASCII tables, ADQL
- Package development and version control with Git
- Computed to development and testing of parallelized Python suite for galaxy image analysis – GalPRIME

Certificates & Courses

- **Parallel Computing School** – ACENET Training Course (*completed 2021*)
- **Job Scheduling with Slurm** – ACENET Training Course (*completed 2021*)
- **Unix Shell, Version Control and Python** – ACENET Training Course (*completed 2021*)
- **Introduction to High-Performance Computing (HPC)** - ACENET Training Course (*completed 2021*)
- **Introduction to Shell Scripting** – ACENET Training Course (*completed 2021*)
- **Visualizing Data with Python** – IBM (*certificate earned August 2020*)
- **Analyzing Data with Python** – IBM (*certificate earned August 2020*)
- **Python Basics for Data Science** – IBM (*certificate earned July 2020*)

Honors & Awards

Research Fellowship (PhD) – Faculty of Graduate Studies & Research (SMU)	2023 – present
Graduate Research Award (PhD) – Faculty of Graduate Studies & Research (SMU)	2023 – present
Best Poster Award (runner-up) – Canadian Astronomical Society's Annual General Meeting	2025
Science Photo Contest Winner – Faculty of Science, Saint Mary's University	2024
Research Fellowship (MSc) – Faculty of Graduate Studies & Research (SMU)	2021 – 2023
Graduate Research Award (MSc) – Faculty of Graduate Studies & Research (SMU)	2021 – 2023
Undergraduate Student Research Award – Natural Sciences & Engineering Research Council of Canada	2021
Academic Achievement Scholarship – Faculty of Science, Saint Mary's University	2019 – 2021
Dean's List – Faculty of Science, Saint Mary's University	2019 – 2021
Edwin Jacob Special University Scholarship in Arts – University of New Brunswick	2012
William and Lois Paine Founder's Scholarship – University of New Brunswick	2012

Leadership & Community Involvement

Chair – Graduate Student Committee, Canadian Astronomical Society (CASCA)	2026-2027
- Represent grad students nationally and serve as their primary liaison with CASCA leadership and committees	
Member, Institute for Computational Astrophysics, Halifax, Canada	2023 – present
- Collaborate with other members, participate in institute-wide research symposiums	
Volunteer Instructor, SHAD Canada, Burke-Gaffney Observatory, Halifax, Canada	07/2025
- Ran several astronomy observation sessions for high school students from across Canada	
Telescope Operator, Solar Eclipse Viewing Event, Burke-Gaffney Observatory	04/2024
- Helped run a solar eclipse viewing event on SMU's campus open to the public	
Social Representative - Residence House Committee, Joy-Kidd House, UNB Fredericton, Canada	2012-2013
- Helped plan & coordinate residence social events and fundraising activities	
Christmas Helper, Grand Bahama Island Toy Drive, Freeport, The Bahamas	2012-2013
- Helped wrap & deliver ~7000 Christmas presents to children across the island of Grand Bahama.	
Youth Soccer Coach, New Brunswick Provincial & Fundy Soccer Programs	2008 – 2010
- Assistant Coach to the U14 Boys NB Provincial Soccer team, and U12 & U14 Boys AAA Fundy Soccer teams	

References

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| - Dr. Ivana Damjanov – ivana.damjanov@smu.ca | Dr. Marcin Sawicki – marcin.sawicki@smu.ca |
| - Dr. Jubee Sohn – jubee.sohn@snu.ac.kr | Dr. Laura Stiles-Clarke – laura.stiles-clarke@smu.ca |